



EXAM PAPERS PRACTICE

Boost your performance and confidence with these topic-based exam questions

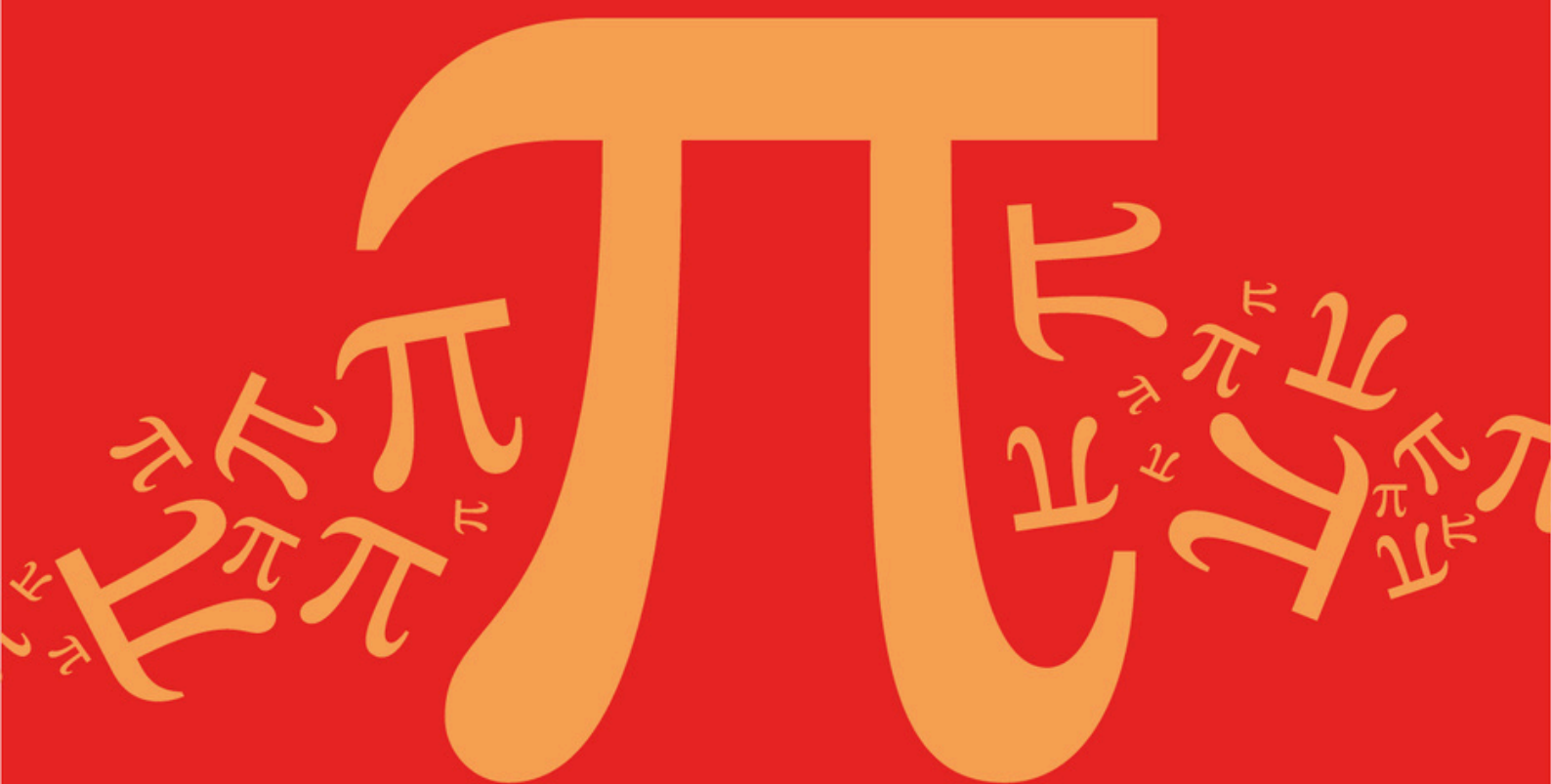
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D15: Graphs of Conics

Mark Scheme

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EXAM PAPERS PRACTICE

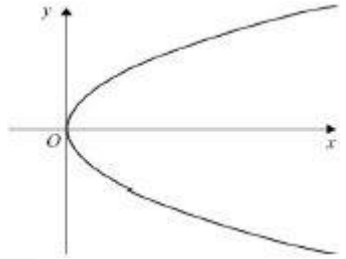
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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Sketches correct parabola	AO1.2	B1	
(b)	Obtains $\pi \int 4x \, dx$ Limits not required for this mark. Condone missing dx	AO3.3	M1	$\text{volume} = \pi \int_0^d y^2 \, dx$ $\therefore 1000 = \pi \int_0^d 4x \, dx$
	Obtains $2x^2$ and uses limits of d and 0 (oe).	AO1.1b	B1	$\frac{1000}{\pi} = \left[\frac{4x^2}{2} \right]_0^d$
	Forms an equation of the form $kd^2 = \text{volume}$ (oe)	AO3.4	M1	$\frac{1000}{\pi} = 2d^2 - 0$ $2\pi d^2 = 1000$
	Correct depth to nearest millimetre. Condone 126 or 12.6 without units. NMS: 126 or 12.6 scores 4/4. Using 1000000 mm ³ leads to a correct answer of 399 mm for 4/4.	AO3.2	A1	$d = \sqrt{\frac{500}{\pi}}$ $\text{depth} = 12.6 \text{ cm}$
	States appropriate assumption	AO3.5b	B1	The thickness of the plastic is negligible
				Total 6 marks

Q2.

Marking Instructions	AO	Marks	Typical Solution
Circles correct answer	AO1.1b	B1	$y = \pm \frac{1}{\sqrt{3}}x$
Total 1 mark			

Q3.

(a) $y = 0, \frac{(x-4)^2}{4} = 1; (x-4)^2 = 4$

OE Sub $y=0$ in eqn of ellipse and either eliminate fraction $\frac{(x-4)}{2} = (\pm) 1$
 or take sq root, condoning missing $\pm$, ie

M1

$$\Rightarrow x = 2, 6 \quad (x_A = 2, x_B = 6)$$

Both 2 and 6

NMS Mark as B2 or B0

A1

2

(b) (i) $\frac{(x-4)^2}{4} + (mx)^2 = 1 \Rightarrow$

Substitute $y = mx$ to eliminate y

M1

$$(x-4)^2 + 4(mx)^2 = 4 \quad x^2 - 8x + 16 + 4(mx)^2 = 4$$

Eliminate fractions correctly and expand $(x-4)^2$ correctly

A1

$$\Rightarrow x^2 - 8x + 16 + 4m^2x^2 - 4 = 0$$

$$\Rightarrow (1 + 4m^2)x^2 - 8x + 12 = 0$$

CSO AG

A1

3

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(ii) Discriminant $b^2 - 4ac \{(-8)^2 - 4(1 + 4m^2)(12)\}$

$b^2 - 4ac$ in terms of m condone one sign or copying error OE

M1

For tangency, $(-8)^2 - 4(1 + 4m^2)(12) = 0$

A correct equation with m^2 being the only unknown at any stage.

A1

$$192m^2 - 16 (=0)$$

OE eg $12m^2 - 1(=0)$ OE PI by a correct value for m condoning wrong sign

A1

($m > 0$ so) $m = \frac{1}{\sqrt{12}}$

ACF of an **exact** value for m

eg $\frac{1}{2\sqrt{3}}, \frac{\sqrt{3}}{6}$, Dep on prev 3 mrks

A1

4

(iii) $(1 + 4 \times \left\{ \frac{1}{\sqrt{12}} \right\}^2) x^2 - 8x + 12 (=0)$

Subst value for m in LHS of eqn (b)(i); ft on c 's value of m .

M1

$$\frac{4}{3}x^2 - 8x + 12 = 0; \quad 4x^2 - 24x + 36 = 0$$

$$x^2 - 6x + 9 = 0$$

$$x = \frac{-(-8) \pm \sqrt{0}}{\frac{8}{3}}; \quad (x - 3)^2 (=0)$$

Valid method to solve a **correct** quadratic equation;
as far as either correct subst into quadratic formula with
 $b^2 - 4ac$ evaluated to 0 or correct factorisation or correct

value of x after $\frac{4}{3}x^2 - 8x + 12 = 0$ or better seen.;

OE, correct use of $-b/2a$

m1

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Must see earlier justification

A1

Coordinates of P are $\left(3, \frac{3}{\sqrt{12}}\right)$

Correct coordinates with the

y -coord in any correct exact form eg $\frac{\sqrt{3}}{2}$

NMS SC 1 for $\left(3, \frac{3}{\sqrt{12}}\right)$

A1

4

[13]

Q4.

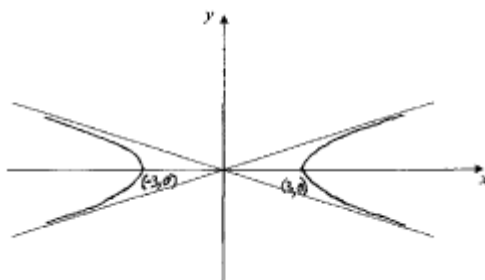
(a) $y = \pm \frac{1}{3}x$

ACF Need both

B1

1

(b)



2-branch curve with branches in correct regions above and below x-axis

B1

Curve approaching asymptotes

B1

Coords $(\pm 3, 0)$, as **only** points of intersection with coordinate axes, indicated. Condone -3 and +3 marked on x-axis at points of intersection as $(\pm 3, 0)$ indicated.

B1

3

(c) (i) $\frac{(x+3)^2}{9} - y^2 = 1$

Replacing x by either $x + 3$ or $x - 3$

M1

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ACF

A1

2

(ii) $\frac{(x+3)^2}{9} - x^2 = 1$

Substitution into c's (c)(i) eqn of $y = x$ to eliminate y or of $x = y$ to eliminate x

M1

$$x^2 + 6x + 9 = 9(x^2 + 1)$$

Correct expansion of $(x \pm 3)^2$ equated to $9(x^2 + 1)$ OE ft; [OE in y]

A1F

$$8x^2 - 6x = 0 \quad (8x^2 = 6x)$$

Ft on error $(x - 3)$ for $(x + 3)$ in (c)(i) which gives $8x^2 + 6x = 0$ ($8x^2 = -6x$) [OE in y]

A1F

Points are $(0,0)$, $(\frac{3}{4}, \frac{3}{4})$
Both. ACF

A1

4

- (d) Adding 3 to c's (c)(ii) **two** x-coords keeping y-coordinates unchanged.

M1

Points are $(3,0)$, $(3\frac{3}{4}, \frac{3}{4})$
*Ft on c's (c)(ii) coordinates for the **two** points*
If not deduced then M0A0

A1F

2

[12]

Q5.

(a) $\begin{bmatrix} c & s \\ -s & c \end{bmatrix}$

B1

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} c & s \\ -s & c \end{bmatrix} \begin{bmatrix} X \\ Y \end{bmatrix}$$

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M1

$$= \begin{bmatrix} cX + sY \\ -sX + cY \end{bmatrix}$$

A1

3

(b) (i) $(cX + sY)^2 - 6(cX + sY)(-sX + cY) - 7(-sX + cY)^2 = 8$

$$(c^2X^2 + 2csXY + s^2Y^2) - 6([c^2 - s^2]XY) + sc\{Y^2 - X^2\} - 7(s^2X^2 - 2csXY + c^2Y^2) = 8$$

Substn. for x & y in eqn. and multiplying out

M1

$$p = c^2 + 6sc - 7s^2$$

A1

$$q = 16cs - 6(c^2 - s^2)$$

AG

A1

$$r = s^2 - 6sc - 7c^2$$

A1

- (ii) Factorising:
 $3s^2 + 8sc - 3c^2 = (3s - c)(s + 3c) = 0$
Or by double angles

M1A1

Deducing a tan value

$$\tan \theta = \frac{1}{3} \quad (\theta \text{ acute})$$

M1

$$\cos \theta = \frac{3}{\sqrt{10}}, \sin \theta = \frac{1}{\sqrt{10}}$$

Both

A1

A1

Substⁿ. sensible values back for p and r

M1

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$$2X^2 - 8Y^2 = 8$$

A1

$$\frac{X^2}{2^2} - \frac{Y^2}{1^2} = 1$$

CSO

A1

8

- (iii) Since C' is a hyperbola, and it is just C rotated, it follows that C is a hyperbola

E1

1

[16]

Q6.

- (a) $(\pm\sqrt{5}, 0), (0, \pm 2)$

If not B2, award B1 if either at least two of these 4 correct

pts or if ' $x = \pm\sqrt{5}$ and $y = \pm 2$ '

B2,1

2

$$(b) \quad \frac{(x-p)^2}{5} + \frac{y^2}{4} = 1$$

Replacing x by either $x + p$ or $x - p$ and keeping y unchanged or as $y \pm 0$

M1

ACF

A1

2

$$(c) \quad \frac{(x-p)^2}{5} + \frac{(x+4)^2}{4} = 1$$

Substitution into c's (b) eqn of $y = x + 4$ to eliminate y

M1

$$4(x-p)^2 + 5(x+4)^2 = 4 \times 5$$

$$4(x^2 - 2px + p^2) + 5(x^2 + 8x + 16) = 20$$

$$4x^2 - 8px + 4p^2 + 5x^2 + 40x + 80 = 20$$

Denominators 5 and 4 cleared in a correct manner and at least either a correct expansion of $(x \pm p)^2$ or a correct expansion of $(x + 4)^2$

m1

$$9x^2 - (8p - 40)x + 4p^2 + 60 = 0$$

CSO No errors in any line of working.

AG. Must see brackets correctly removed and all terms involving x , p correctly rearranged to same side before the printed answer is stated. Must have ' $= 0$ ' although brackets around $4p^2 + 60$ may be omitted

A1

3

$$(d) \quad \text{Discriminant is } (8p - 40)^2 - 4(9)(4p^2 + 60)$$

OE Must be isolated, not just within the quadratic formula

B1

For tangency

$$(8p - 40)^2 - 4(9)(4p^2 + 60) = 0$$

OE Equating c's discriminant to zero before obtaining any values for p

M1

$$p^2 + 8p + 7 = 0$$

ACF with like terms collected

A1

$$\{(p + 1)(p + 7) = 0 \Rightarrow \} p = -1, p = -7(*)$$

Correct values -1, -7 for p

B1

$$p = -1: 9x^2 + 48x + 64 (= 0)$$

$$p = -7: 9x^2 + 96x + 256 (= 0)$$

Substitutes at least one of c's **two** values for p either

into the given quadratic in (c) OE or into $\frac{8p - 40}{18}$

M1

$$p = -1: 9x^2 + 48x + 64 (= 0) \Rightarrow x = -\frac{8}{3}$$

$x = -\frac{8}{3}$ OE as only root from the quadratic or from $\frac{8p - 40}{18}$. Apply FIW if (*) is B0

A1

$$p = -7: 9x^2 + 96x + 256 (= 0) \Rightarrow x = -\frac{16}{3}$$

$x = -\frac{16}{3}$ OE as only root from the quadratic or from $\frac{8p - 40}{18}$. Apply FIW if. (*) is B0

A1

$$x = -\frac{8}{3}, y = \frac{4}{3}, x = -\frac{16}{3}, y = -\frac{4}{3}$$

$$\left(-\frac{8}{3}, \frac{4}{3}\right) \quad \left(-\frac{16}{3}, -\frac{4}{3}\right)$$

CSO Previous 7 marks must have been awarded and coordinates of both points need to be correct and exact but accept in either format

A1

8

[15]

Q7.

- (a) Sketch of ellipse
centred at origin

M1

Correct relationship to circle

A1

Coords $(\pm 2\sqrt{2}, 0)$ $(0, \pm \sqrt{2})$

Accept $\sqrt{8}$ for $2\sqrt{2}$;

B1 for any 2 of $x = \pm 2\sqrt{2}$, $y = \pm \sqrt{2}$

allow B1 if all correct except for use of decimals (at least one DP)

B2,1

4

- (b) (i) Replacing x by $\frac{x}{2}$
or by $2x$

M1

E is $\left(\frac{x}{2}\right)^2 + y^2 = 2$

OE

A1

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2

- (ii) Tangent is $\frac{x}{2} + y = 2$

M1 for complete valid method

M1A1

2

[8]

Q8.

- (a) (i) Elimination to give $x = \frac{1}{8}x^2$
OE

M1

A is (8, 8)

NMS 2/2

A1

2

- (ii) Equation of Q is $x = \frac{1}{8}y^2$
OE; condone $y = \sqrt{8x}$

B1

1

- (iii) Points of contact are images in $y = x$

E1

1

- (b) (i) Eliminating y to give $-x + c = \frac{1}{8}x^2$

M1

(ie $x^2 + 8x - 8c = 0$)

Distinct roots if $\Delta > 0$
stated or implied

E1

$\Delta = 64 + 32c$, so $c > -2$
convincingly shown (AG)

A1

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- (ii) For tangent $c = -2$, so $x^2 + 8x + 16 = 0$
OE

M1

... and $x = -4$, $y = 2$

A1

Reflection in $y = x$
or other complete method

M1

$x = 2$, $y = -4$
ft wrong answer for first point; allow NMS 2/2

A1F

4

[11]

Q9.

(a) $x = 2, y = 0 \Rightarrow \frac{4}{a^2} - 0 = 1$ so $a = 0$
E1 for verif'n or incomplete proof

E2,1

Asymps $\Rightarrow \pm \frac{b}{a} = \pm 2$ so $b = 2a = 4$
ditto

E2,1

4

(b) Line is $y - 0 = m(x - 1)$
OE

B1

Elimination of y

M1

$4x^2 - m^2(x^2 - 2x + 1) = 16$
OE (no fractions)

A1

So $(m^2 - 4)x^2 - 2m^2x + (m^2 + 16) = 0$
convincingly shown (AG)

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A1

4

(c) Discriminant equated to zero

M1

$4m^4 - 4m^4 - 64m^2 + 16m^2 + 256 = 0$
OE

A1

$-3m^2 + 16 = 0$, hence result
convincingly shown (AG)

A1

3

(d) $m^2 = \frac{16}{3} \Rightarrow \frac{4}{3}x^2 - \frac{32}{3}x + \frac{64}{3} = 0$

M1

$$x^2 - 8x + 16 = 0, \text{ so } x = 4$$

m1A1

Method for y-coordinates

$$m = \pm \frac{4}{\sqrt{3}}$$

using $\frac{4}{\sqrt{3}}$ or from equation of hyperbola; dep't on previous m1

m1

$$y = \pm 4\sqrt{3}$$

A1

5

[16]

Q10.

- (a) (i) Parabola drawn
with x-axis as line of symmetry

M1

passing through (2, 0)

A1

2

- (ii) Two tangents passing through (−2, 0)
to c's parabola

B1B1

2

- (b) (i) Elimination of y

M1

Correct expansion of $(x + 2)^2$

B1

Result

convincingly shown (AG)

A1

3

- (ii) Correct discriminant

B1

$$16m^4 - 8m^2 + 1 = 16m^4 + 8m^2$$

OE

M1

Result

convincingly shown (AG)

A1

3

(iii) $\frac{1}{16}x^2 - \frac{3}{4}x + \frac{9}{4} = 0$

OE

M1

$x = 6, y = \pm 2$

A1,A1

3

[13]

Q11.

- (a) Asymptotes are $y = \pm \sqrt{2}x$

M1A0 if correct but not in required form

M1A1

2

- (b) Asymptotes correct on sketch

*With gradients steeper than 1;
ft from $y = \pm mx$ with $m > 1$*

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B1F

Two branches in roughly correct positions

B1

Approaching asymptotes correctly

Asymptotes $y = \pm mx$ needed here

B1

3

- (c) (i) Elimination of y

M1

Clearing denominator correctly

M1

$x^2 - 2cx - (c^2 + 2) = 0$

Convincingly found (AG)

m1A1

4

- (ii) Discriminant = $8c^2 + 8$
 $\dots > 0$ for all c , hence result
Accept unsimplified

B1

OE

E1

2

- (iii) Solving gives $x = c \pm \sqrt{2(c^2 + 1)}$

M1A1

$$y = x + c = 2c \pm \sqrt{2(c^2 + 1)}$$

Accept $y = c + \frac{2c \pm \sqrt{8c^2 + 8}}{2}$

A1

3

[14]

EXAM PAPERS PRACTICE

Q12.

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- (a) (i)

$$\frac{1}{5} \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$$

1/det

B1

Transposed matx. of cofactors

B1

2

- (ii) $\begin{bmatrix} x \\ y \end{bmatrix} = \mathbf{A}^{-1} \begin{bmatrix} X \\ Y \end{bmatrix} = \begin{bmatrix} \frac{1}{5}(X + 2Y) \\ \frac{1}{5}(Y - 2X) \end{bmatrix}$

M1

ft

A1F

2

(b) $\mathbf{A} = \sqrt{5} \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}$

B1

Enlargement sf $\sqrt{5}$ (centre O)
Two components in any order

M1A1

+ Rotation thro' $\cos^{-1}(1/\sqrt{5})$
or 63.4° or 1.11 rads

M1A1

5

(c) (i) $p = \frac{1}{2}$, $q = 3$ noted

Or form $\frac{\frac{x^2}{1}}{\frac{1}{2}} + \frac{\frac{y^2}{3}}{\frac{1}{3}} = 1$ shown

B1

1

(ii) $6x^2 + y^2 = 3 \Rightarrow$

$$\frac{6}{25}(X^2 + 4XY + 4Y^2) + \frac{1}{25}(Y^2 - 4XY + 4X^2) = 3$$

Substⁿ. for x and y

M1

$$\Rightarrow 10X^2 + 20XY + 25Y^2 = 75$$

$$\Rightarrow 2X^2 + 4XY + 5Y^2 = 15$$

ANSWER GIVEN

A1

2

(iii) It is just an enlarged rotation of E ,
 hence still an ellipse

B1

1

[13]

Q13.

(a) Ellipse with centre of origin
Allow unequal scales on axes

B1

$(\pm \sqrt{3}, 0)$ and (0 ± 2) shown on diagram

Condone AWRT 1.7 for $\sqrt{3}$;

B1 for incomplete attempt

B2,1

3

(b) y replaced by $\frac{1}{2}y$

M1A0 for $2y$ instead of $\frac{1}{2}y$

M1A1

Equation is now $\frac{x^2}{3} + \frac{y^2}{16} = 1$

A1

3

(c) Attempt at completing the square

$$4(x-1)^2 + 3(y+1)^2 \dots$$

M1

A1A1

[Alt: replace x by $x-a$ and y by $y-b$

M1 if one replacement correct

(M1)

$$4x^2 - 8ax + 3y^2 - 6by \dots$$

Condone errors in constant terms

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(m1A1)

$$a = 1 \text{ and } b = -1$$

A1A1

5

[11]

Q14.

(a) Asymptotes are $y = \pm \frac{1}{2}x$

OE; M1 for $y = \pm mx$

M1A1

2

(b) $x = 4$ substituted into equation

M1

$$y^2 = 3 \text{ so } y = \pm \sqrt{3}$$

Allow NMS

A1

2

- (c) (i) y -coords are $2 \pm \sqrt{3}$
ft wrong answer to (b)

B1F

1

- (ii) Hyperbola is $\frac{x^2}{4} - (y - 2)^2 = 1$
M1A0 if $y + 2$ used

Asymptotes are $y = 2 \pm \frac{1}{2}x$
ft wrong gradients in (a)

M1A1

B1F

3

[8]

Q15.

- (a) Curve translated 7 in y direction

B1

... and 1 in negative x direction
or answer in vector form

B1

2

- (b) (i) Asymptotes $x = -1$ and $y = 7$

B1B1

2

- (ii) Intersections at $(0, 8)$...

B1

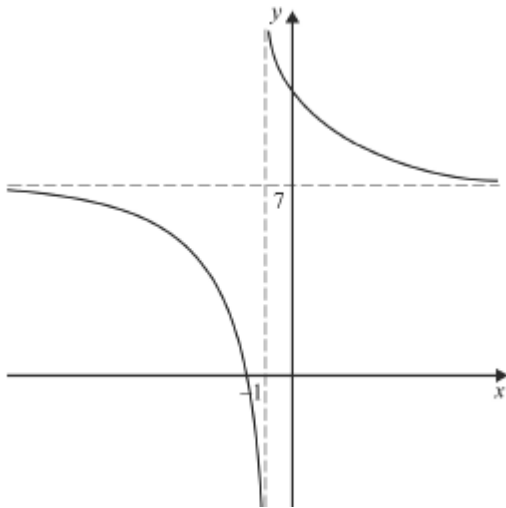
... and $(-\frac{8}{7}, 0)$

Allow AWRT -1.14 ; NMS $1/2$

M1A1

3

- (c)



At least one branch
of correct shape

B1

Complete graph
translation of $y = 1/x$

B1

All correct including asymptotes
in roughly correct positions

B1

3

[10]

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Q16.

- (a) Equation is $y - 4 = m(x - 3)$
OE; M1A0 if one small error

M1A1

2

- (b) Elimination of x

M1

$$4y - 16 = m(y^2 - 12)$$

OE (no fractions)

A1

Hence result
convincingly shown (AG)

A1

3

- (c) Discriminant equated to zero

M1

$$(3m - 1)(m - 1) = 0$$

OE; m1 for attempt at solving

m1A1

Tangents $y = x + 1, y = \frac{1}{3}x + 3$

OE

A1A1

5

(d) $m = 1 \Rightarrow y^2 - 4y + 4 = 0$

OE; $m = 1$ needed for this

M1

so point of contact is (1, 2)

A1

$$m = \frac{1}{3} \Rightarrow \frac{1}{3}y^2 - 4y + 12 = 0$$

OE; $m = \frac{1}{3}$ needed for this

M1

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so point of contact is (9, 6)

A1

4

[14]

Q17.

(a) $x = 10 \Rightarrow 4 - \frac{y^2}{9} = 1$

M1

$$\Rightarrow y^2 = 27$$

PI

A1

$$\Rightarrow y = \pm 3\sqrt{3}$$

A1

3

- (b) One branch generally correct
Asymptotes not needed

B1

Both branches correct
With implied asymptotes

B1

Intersections at $(\pm 5, 0)$

B1

3

- (c) Required tangent is $x = 5$
ft wrong value in (b)

B1F

1

- (d) (i) y correctly eliminated

M1

Fractions correctly cleared

m1

$$16x^2 - 200x + 625 = 0$$

convincingly shown (AG)

A1

3

(ii) $x = \frac{25}{4}$

*No need to mention repeated root,
 but B0 if other values given as well*

B1

Equal roots $\Rightarrow$ tangency
Accept 'It's a tangent'

E1

2

[12]

Q18.

- (a) Intersections $(\pm\sqrt{2}, 0)$, $(0, \pm 1)$

Allow B1 for $(\sqrt{2}, 0)$, $(0, 1)$

B1B1

2

- (b) Equation is $\frac{(x-k)^2}{2} + y^2 = 1$
M1 if only one small error, eg $x + k$ for $x - k$

M1A1

2

- (c) Correct elimination of y

M1

Correct expansion of squares

M1

Correct removal of denominator

M1

Answer convincingly established

AG

A1

4

- (d) Tgt $\Rightarrow 4(k+4)^2 - 12(k^2+6) = 0$

M1

.... $\Rightarrow k^2 - 4k + 1 = 0$

OE

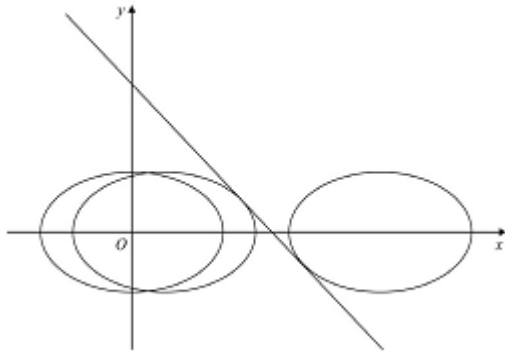
m1A1

... $\Rightarrow k = 2 \pm \sqrt{3}$

A1

4

- (e)



Curve to left of line

B1

Curve to right of line

B2

3

Curves must touch the line in approx correct positions
 SC 1/3 if both curves are incomplete but touch the line correctly

[15]

Q19.

- (a) Good attempt at sketch

M1

Correct at origin

A1

2

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- (b) (i) y replaced by $y - 2$

B1

Equation is $(y - 2)^2 = 12x$
 ft $y + 2$ for $y - 2$

B1ft

2

- (ii) Equation is $x^2 = 12y$

B1

1

- (c) (i) $(x + c)^2 = x^2 + 2cx + c^2$

B1

... = $12x$

M1

Hence result
convincingly shown (AG)

A1
3

(ii) Tangent if $(2c - 12)^2 - 4c^2 = 0$

M1

ie if $-48c + 144 = 0$ so $c = 3$

A1
2

(iii) $x^2 - 6x + 9 = 0$

M1

$x = 3, y = 6$

A1
2

(iv) $c = 4 \Rightarrow$ discriminant $= -48 < 0$
OE

M1A1

So line does not intersect curve

A1
3

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[15]

Q20.

(a) Stretch parallel to y axis ...

B1

... scale-factor $\frac{1}{2}$ parallel to y axis

B1
2

(b) $(x - 2)^2 - y^2 = 1$

M1A1

Translation in x direction ...

A1

... 2 units in positive x direction

A1

4

[6]

Q21.

(a) Correct shape

Coordinates $(\pm 3, 0)$, $(0, \pm 2)$ shown

B1

Allow labels on sketch

B2,1

3

(b) Attempt to solve $\frac{1}{9} + \frac{y^2}{4} = 1$

M1

At least one correct root

Allow decimals; allow $\sqrt{\frac{32}{9}}$

m1

$$y = \pm \frac{4}{3}\sqrt{2}$$

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A1

3

(c) Eqn is $\frac{(x-1)^2}{9} + \frac{y^2}{4} = 1$

M1A0 for eg wrong sign

M1A1

2

[8]

Examiner reports

Q1.

Most students drew a correct sketch of the curve, although some drew only the top half.

In part (b), the majority of students used centimetres and produced a bowl with sensible dimensions. This question was generally well answered, although a significant minority of students initially included π , but then lost it during their calculations.

In part (c), many students gave a sensible assumption, usually referring to the thickness of the material used. Some incorrect responses questioned the capabilities of 3-D printers.

Q3.

This final question was again answered correctly by the majority of students. Most students were able to find the correct x -coordinates for A and B , some by considering a

translation of the 'basic' ellipse $\frac{x^2}{4} + y^2 = 1$. In part (b)(i) most students showed sufficient care in reaching the printed equation, but it was also surprising to see some final answers which contained sign errors and did not match the printed answer. Most students used the discriminant in a correct manner although a significant minority gave two values for m , not noting the given information in the first line of the question for (b) that m was positive. Less able students generally made little progress with the final part of the question, but most other students obtained the correct coordinates for P .

Q4.

This question, involving hyperbolas, was generally answered well although the final part was frequently not answered in the required manner.

In part (a) some gave the wrong equations $y = \pm 3x$ for the asymptotes and some others did not even give an equation.

Most candidates, in part (b), were able to show the main features of the curve in their sketches and to state the correct coordinates of the points of intersection with the axis although some other sketches were seen which bore little resemblance to the shape of a hyperbola.

As expected, the most common wrong answer for part (c)(i) was ' $\frac{(x-3)^2}{9} - y^2 = 1$ ' but this error was not overly penalised as some follow through was allowed in the later part. There were, however, many correct solutions to both parts (c)(i) and (c)(ii).

Generally only the more able candidates were able to **deduce** the correct coordinates in part (d). Candidates who started afresh, making no reference to their answers in part (c)(ii), were awarded no credit. Common wrong follow through answers were obtained by either subtracting 3 from their x -coordinates in part (c)(ii) or adding / subtracting 3 to / from their y -coordinates in part (c)(ii).

Q5.

Although there were a few excellent, complete responses to this question, in general it turned out to reflect the lack of preparedness on the part of candidates for the demands

presented to them here. The three marks in part (a) were readily and easily acquired by most candidates attempting it. However, 156 candidates were unable to proceed beyond the given result of part (a), and made no attempt at the remaining parts of this question.

In part (b)(i), it was not uncommon to find that candidates were less than comfortable with squaring out their x and y terms in the new variables X and Y , and collecting them up proved to be far more problematical than should have been the case. Rather a lot of candidates seemed to think that the XY term came exclusively from the xy term, and such efforts foundered at this point.

Few candidates got to the correct equation of the hyperbola. A suitable answer to part (b)(iii) required only an awareness that the original curve had been rotated into the final hyperbola, but only a few candidates answered it without having worked their way to it. Of these, most offered spurious reasons, such as the not uncommon reference to the suggestion that the original curve was a hyperbola because this final one was an ellipse.

Q6.

The majority of students had a clear understanding of the methods required to answer this final question which was based on the equation and translation of an ellipse. In part (a) most students were able to find the values of x when $y = 0$ and the values of y when $x = 0$ but a significant minority of students either gave the coordinates incorrectly or did not attempt to express the values in coordinate form. In part (b), the majority of students gave

the correct equation of the translated ellipse but the incorrect answers $\frac{(x+p)^2}{5} + \frac{y^2}{4} = 1$

and $\frac{x^2 - p^2}{5} + \frac{y^2}{4} = 1$ were also seen. In part (c), the correct substitution of $y = x + 4$ was generally used and most indicated that they knew how to clear brackets and fractions but in attempting to reach the printed result, algebraic working was not always absolutely correct. In such questions, each line of the student's solution has to be correct and any corrections made must be made to all relevant lines of working. In part (d), the majority of students correctly equated their discriminant to zero although some inequalities were also seen. Those who used $(40 - p)$ as their value for ' b ' were slightly more successful than those who used $-(p - 40)$. Those who failed to arrive at the correct quadratic equation in p , which they then tried to solve, often knew they had to substitute their value(s) into the given quadratic in part (c) but could go no further. A minority of students however, having obtained the correct equation $p^2 + 8p + 7 = 0$ then surprisingly gave the x -coordinates as -1 and -7 . Although sign errors were sometimes seen in substituting -1 and -7 for p into the quadratic in part (c), the majority of students who obtained the correct quadratic equations in x usually went on to find the correct coordinates of the points of contact.

Q7.

In part (a) of this question, most candidates managed to draw an acceptable attempt at an ellipse touching the given circle in the appropriate places. Occasionally the stretch would be applied parallel to the y -axis rather than the x -axis. The required coordinates were indicated with various levels of accuracy, sometimes appearing as integers, sometimes with minus signs omitted.

In part (b), the most successful candidates were often the ones who wrote the least — all that

was needed was to replace x by $\frac{x}{2}$ in each of the two given equations. Many candidates answered part (b)(i) concisely and correctly but then went into a variety of long methods to find the equation of the tangent in part (b)(ii). Some used implicit differentiation and were

usually successful. Others used chain-rule differentiation and usually made errors. Yet others embarked on a very complicated piece of work based on quadratic theory and using a general gradient m . This must have consumed a large amount of time and was almost invariably unsuccessful.

Q8.

In part (a)(i), some candidates found the short method of using the equation $y = x$ to find the coordinates of A . More commonly, the equation of the parabola Q was used in conjunction with that of P , and after some relatively heavy algebra the required coordinates were found correctly. Most candidates were able to write down the equation of Q in part (a)(ii), or indeed had already written it down in their working for the first part of the question. Only a handful of candidates gave an adequate answer in part (a)(iii). Many stated that the common tangent had to be perpendicular to the mirror line, but for this to be entirely convincing it was necessary to mention the two points of contact of the line with the parabolas, which must be mirror images of each other.

Part (b)(i) gave candidates the opportunity to show their knowledge of quadratic theory, though in some cases they failed to keep to the rules for correct manipulation of inequalities. In part (b)(ii), some candidates immediately saw that c must be equal to -2 for the tangency case, while others virtually repeated their work done in the previous part, with equals signs, to establish the correct value of c . Once the coordinates of one point of contact had been found, it was possible to write down the other pair by using a reflection in $y = x$, but many candidates performed another calculation to find the second pair of coordinates.

Q9.

Part (a) was found very hard by most candidates. Many failed to use both pieces of information supplied just before part (a), so that they could establish $a = 2$ or $b = 2a$ but could not hope to complete the two requests. Whether they were attempting one half or both halves of the question, they often wrote down the results they were supposed to be proving, possibly earning some credit for verifying these results, though the reasoning was sometimes very hard to follow. Part (b) was much more familiar to well-prepared candidates, but marks were often lost either by a failure to form a correct equation for the straight line or by sign errors after the elimination of y . The solutions to parts (c) and (d) were often presented in the reverse order, but full credit was given for all correct working shown. In part (c), many candidates made a good attempt to deal with the discriminant of the quadratic equation printed in part (b), but were careless about indicating that this discriminant should be equal to zero for equal roots. Once again, sign errors often caused a loss of marks. In part (d), the unique value of x was often found correctly by the stronger candidates, but relatively few of these went on to find the values of y , and those who did sometimes did so via a rather roundabout approach.

Q10.

The sketch of the parabola P was generally well attempted. When asked to sketch two tangents to this parabola, many candidates revealed a poor understanding of the idea of a tangent to a curve.

Part (b) was found familiar by all good candidates, and parts (b)(i) and (b)(ii) yielded full marks provided that a little care was taken to avoid sign errors. Part (b)(iii) was more demanding but many candidates found their way to earning at least some credit, either by substituting the value of m^2 into the quadratic found in part (b)(i) or by some more

roundabout method.

Q11.

As in the June 2008 paper, there were many candidates who appeared to struggle with the earlier parts of the final question but then came into their own when they reached the expected test of quadratic theory. Although the asymptotes of a hyperbola had appeared before on MFP1 papers, many candidates seemed unfamiliar with them, and even with the hyperbola itself. A common error in part (a) was to fail to take a square root. In part (b), even those who drew the asymptotes and the two branches in roughly the right positions rarely showed the curve actually approaching the supposed asymptotes.

Part (c)(i) was extremely well answered, the printed answer guiding most candidates unerringly towards a correct piece of algebra, though some made a sign error and then made another one in order to reach the answer. The minus signs caused many candidates to go wrong in finding the discriminant for part (c)(ii), and the fact that the discriminant needed to be strictly positive was not stated in many cases.

The simplest approach to part (c)(iii) was to solve the given equation for x in terms of c and then to add another c to obtain the values of y , but most candidates did not seem to realise this. Many candidates obtained a valid quadratic equation for y but still failed to write down the quadratic formula as applied to their equation, and thus failed to earn any marks in this part. Those who attempted to substitute values into the formula, either for x or for y , often omitted the factor c from the coefficient of x and/or ignored the minus sign in front of this coefficient.

Q12.

It was disappointing to see so many candidates unable to find correctly the inverse of a 2×2 matrix at the outset of this question. The description in part (b) proved tricky — as had been expected — but actually only relies on understanding of the FP1 work on matrices and transformations. Taking this into account, it was sad to see so few attempts to describe T .

In part (c)(i), many candidates failed to point out the values of p and q that would justify E 's status as an ellipse. In most cases, this arose as a result of an inability to divide correctly throughout the given equation by 3. By the time it came to the final two parts, most candidates had made at least one mistake which deprived them of the chance of further successful progress.

Q13.

The sketches of the ellipse in part (a) of this question were mostly satisfactory, though some candidates failed to take square roots when working out the required coordinates.

Part (b) was one of those parts of questions where the absence of any explanation made it hard for the examiners to see what the candidates were trying to do. Some gave the answer in simplified form without any preliminary working. On this occasion the examiners condoned this poor examination technique where it looked as if the candidate had done the right thing. But there were many errors here: some modified the x term rather than the y term, some multiplied the y by 2 instead of dividing it by 2, and many failed to include this 2 in the squaring process when simplifying their equations.

Part (c) was found to be rather hard, but many candidates realised that completing the square was needed and at least made an effort, often failing to see the connection

between their correct equations and the required numbers a and b . The alternative approach of applying the translation from the outset, using the letters a and b , was about as popular as the completing the square method, but the algebra became too heavy for many candidates and the necessary comparing of terms did not take place.

Q14.

The oblique asymptotes of a hyperbola were not always known by the candidates, though many found the necessary general equations in the formula booklet and correctly applied them to this particular case. Part (b) was very well answered by almost all candidates, while part (c) usually provided some further marks, the most common error being to use $y + 2$ instead of $y - 2$ in the equations of the translated hyperbola and asymptotes.

Q15.

Many candidates scored well on this question but, for some, marks were lost in a variety of places.

In part (a) some candidates were careless in giving the two parts of the translation.

In part (b) (i) the horizontal asymptote was often found to be the x -axis, which usually caused difficulty in part (c). In part (b)(ii) a mark was sometimes lost by a failure to show some necessary working, in this case for the intersection of the curve with the x -axis.

In part (c) the sketches were sometimes wildly wrong, despite the information provided in part (a) that the curve must be a translated version of the well-known hyperbola $y = \frac{1}{x}$. In many cases the curve shown was basically correct but did not appear to approach the asymptotes in a satisfactory way. Despite a generous interpretation of this on the part of the examiners, some candidates lost credit because of seriously faulty drawing.

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Q16.

Most candidates know that there is likely to be a question involving quadratic theory and are well equipped to answer it. In this case it was possible to answer parts (c) and (d) without having been successful in the earlier parts of the question, and this was often seen. At the same time there were many candidates who did well in parts (a) and (b) but made an error in the discriminant in part (c) leading to a significant loss of marks thereafter.

In part (a) many candidates found a particular value for m rather than finding an equation that would be valid for all values of m . In part (b), as mentioned above, the elimination was not always carried out by the most efficient method, but many candidates still managed to establish the required equation. In part (c) some candidates lost all credit by failing to indicate that the discriminant must be zero for the line to be a tangent to the parabola; while others found the two gradients correctly but omitted the actual equations asked for in the question.

Those who did find the correct gradients nearly always went on to gain all or most of the marks available in part (d), from a quadratic in y (the more direct way) or from a quadratic in x .

Q17.

Part (a) of this question was generally well answered apart from the omission of the plus-or- minus symbol by a sizeable minority of candidates. The wording of the question should have made it very clear that there would be more than one point of intersection.

In part (b) many candidates sketched an ellipse instead of a hyperbola. Those who realised that it should be a hyperbola often lost a mark by showing too much curvature in the parts where the curve should be approaching its asymptotes.

Most candidates gave an appropriate answer (in the light of their graph) to part (c). No credit was given here to those who had drawn an incorrect curve, such as an ellipse, which just happened to have a vertical tangent at its intersection with the positive x -axis.

It was good to see that, no doubt helped by the printed answer, the majority of candidates coped successfully with the clearing of fractions needed in part (d)(i). Some candidates seemed to ignore the instruction to solve the equation in part (d)(ii) and went straight on to the statement that the line must be a tangent to the curve. Others mentioned the equal roots of the equation but failed to mention any relationship between the line and the curve.

Q18.

This final question presented a varied challenge, mostly focused on the algebraic skills needed to cope with the quadratic equation printed in part (c) of the question – deriving this equation from two other equations and then using the discriminant of the equation to find the cases where the ellipse would touch the line. Many candidates were well prepared for this type of question and scored heavily, though with occasional errors and omissions.

Part (e) of the question was often not attempted at all, or the attempts were totally incorrect, involving transformations other than translations parallel to the x -axis. In some cases only one of the two possible cases was illustrated, usually the one where the ellipse touched the straight line on the lower left side of the line, but an impressive minority of scripts ended with an accurate portrayal of both cases.

Q19.

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Marks were often lost in part (a) by a sketch which showed only the upper part of the parabola, or by the appearance of a cusp at the origin instead of a smooth curve touching the y -axis.

Part (b)(i) was also answered incorrectly by many candidates, who inserted $+ 2$ or $- 2$ in inappropriate places. The transformation in part (b)(ii) was better known.

In part (c) many candidates sailed through all the parts in good style and picked up ten marks out of ten.

Part (c)(i) was a particularly straightforward test for almost all the candidates. In part (c)(ii) a common mistake was to give the discriminant as $(2c - 12)^2 - 4$, ignoring the c^2 which was the constant term of the quadratic in part (c)(i). Some candidates omitted the y -coordinate in part (c)(iii), or gave it as ± 6 . Part (c)(iv) provided most candidates with the opportunity to end on a high note with a brief but correct response.

Q20.

In order to gain full marks in this question it was necessary to use the correct standard technical words 'stretch' and 'translation'. In part (a) most candidates did indeed use the word 'stretch', but very rarely with the correct scale factor. The candidates were not always familiar with the process of completing the square, but when they did carry out this

requirement correctly they usually went on to give the correct transformation in part (b).

Q21.

Part (a) was usually answered correctly, although some candidates seemed to spend more time on it than they would have had if they had been more familiar with the context. Weaker candidates struggled with the algebra in part (b), but the more confident ones missed the significance of the plurals in the wording of the question and found only one y -coordinate. Most candidates wrote down their answers in part (c) as required, often correctly but sometimes with an extraordinary variety of errors.



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